Partner-Specific Affective Precision in Social Active Inference

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Abstract. Beyond its role in shaping subjective experience, affect regulates how decisively beliefs are expressed in action. It can be understood as a structural component of intelligent behavior in uncertain social environments: a signal that assigns action-relevant weight to beliefs, determining the degree of commitment rather than the content of inference. The active inference framework formalizes this computationally, modeling affect as a metacognitive signal through a running estimate of generative model fitness that modulates action confidence via an inferred precision parameter. Social confidence, however, is inherently relational, earned through each partner's specific behavioral history. We extend this formulation to multi-partner social settings, giving each social relationship its own model-fitness estimate and action-confidence budget updated from that partner's evidence alone.
Simulations in a multi-partner trust game demonstrate that partner-local affective precision tracks predictability rather than value: a reliably hostile and a reliably cooperative partner can both earn high precision, because what they share is predictability, not valence. In decision-open regimes, deployment changes arise through action sharpening rather than improved belief quality; in partner-choice settings, preliminary allocation readouts suggest that engagement begins to reorganize around reliability before aggregate payoff separates. Under abrupt behavioral change, precision can improve payoff even when type accuracy falls, indicating adaptive reallocation across the social portfolio rather than superior inference about the changed partner. This exposes a temporal dependency: accumulated confidence is useful when it can be redirected toward reliable alternatives quickly enough, but it remains sensitive to the structure of available relationships and the speed of change. Varying precision-gain and the prior over model fitness defines computational profiles connecting parameterized confidence dynamics to empirically observed variation in social trust. Affect, in this account, is a metacognitive signal whose dynamics and individual variation reflect the temporal and relational structure of social life.

Keywords: Active inference · Affect · Social cognition · Multi-agent systems · Trust · Individual differences

1 Introduction

Social cognition requires two things that formal models rarely separate. The first is prediction: forming expectations about what others know, intend, and are likely to do. Theory-of-mind accounts and computational models of social inference address this with increasing sophistication [10, 11]. However, beyond accurate prediction, social behavior also requires committing: deciding how decisively to act on what you know. Knowing that someone is unreliable is not the same as knowing how cautiously to treat them. Most formal accounts address the first problem while leaving the second implicit.

In psychology and neuroscience, affect refers to the valence and arousal that gives experience its motivational texture—the felt pull toward or away from situations, relationships, and social commitments [2]. It is most often treated as a descriptor of inner states, but affect can also mediate the commitment problem. Rather than merely coloring experience, it regulates how strongly beliefs drive behavior, assigning action-relevant weight to what an agent already infers and determining the degree of commitment rather than the content of inference. An agent that infers accurately but commits poorly wastes its social knowledge; one that commits confidently on fragile models exposes itself to exploitation.

The active inference framework [7, 8, 16] provides a formal route from this observation to a computational mechanism. Agents minimize variational free energy to update beliefs and evaluate candidate policies by their expected free energy [9, 6]. Action selection is precision-weighted: the same beliefs can produce decisive or tentative behavior depending on γ, the parameter controlling how sharply the agent commits to its most-preferred policy. Hesp et al. [13] made affect endogenous by modeling it as active inference over a higher-order belief β representing subjective model fitness. An affective charge signal ϕ drives updates to β, rising when the model performs better than a prior baseline and falling when it performs worse, making affect a derivative of model fitness rather than reward history.

This affect-as-model-fitness account has not, to our knowledge, been extended to relationship-specific confidence in multi-partner social settings, and the extension raises a problem that does not arise in single-agent formulations. Social confidence is inherently relational. In daily social life, we do not carry the same felt certainty toward every person we know—we carry it relationship by relationship, earned through each partner's specific behavioral history. A person can hold a well-validated model of a longtime colleague, a fragile model of a new acquaintance, and an actively revising model of someone who has recently surprised them; these three states of confidence should not be pooled into one shared precision budget. We therefore extend affective precision to multi-partner social settings, giving each social relationship its own model-fitness estimate and action-confidence budget updated from that partner's behavioral evidence alone.

We test the dynamics this produces in a repeated multi-partner trust game across binary and graded decision regimes, partner-choice conditions, abrupt betrayal, and phenotype-style parameter perturbations. The results characterize both the functional role and the temporal boundary conditions of relationship-specific social confidence. The paper tests three linked claims: first, partner-local affective precision tracks model reliability rather than realized payoff; second, its behavioral effects arise from action confidence rather than improved partner-state beliefs; and third, the dynamics of precision gain and prior model fitness define temporally structured profiles of social trust revision.

2 Methods

A focal active-inference agent plays a repeated trust game with four partners whose types and stances are hidden and must be inferred from behavioral evidence. Rather than maintaining a single pooled social model, the agent keeps a separate generative model for each partner, allowing evidence to accumulate independently across relationships. Partner-local affective precision is then layered on top: each partner has its own model-fitness estimate that modulates how confidently the agent acts on that partner's model.

2.1 Trust game

The trust game [4, 14] is a natural testbed because partner types are hidden and must be inferred, creating a genuine gap between social knowledge and actionable confidence. The policy space can also be varied: binary settings offer two actions, while graded settings distribute probability mass across several investment levels, and a partner-choice regime makes approach and avoidance explicit policy outputs.

Each partner has a latent *type* (cooperator, reciprocator, exploiter, or random) and a latent *stance* toward the agent (trusting, neutral, or hostile)—neither directly observable; both must be inferred from behavioral evidence. Episodes run for up to 200 rounds.

In the binary game both players choose to cooperate or defect. Table 1 gives the focal agent's payoffs.

Table 1: Focal agent payoffs in the binary trust game.

	Partner cooperates	Partner defects
Agent cooperates	3	-1
Agent defects	5	1

The structure poses a social inference problem: defecting is individually tempting regardless of partner behavior, but a cooperative partner is worth engaging, so the agent must infer type and stance to choose well. In the graded regime, the binary choice is replaced by a discretized investment level; the agent commits a resource along a continuous scale rather than making a binary commitment. In the partner-choice regime, the agent selects which partner to engage before choosing an action each round.

2.2 Per-partner generative models

Given this environment, the focal agent maintains a separate discrete generative model $\mathcal{M}_k$ for each partner k, rather than a single pooled social model. This keeps evidence about each partner's behavior independent and allows the confidence with which each model is acted upon to differ across partners. Each model encodes observation likelihoods (A_k^{act}, A_k^{pay}), transition dynamics (B_k), preferred observations (C_k), prior beliefs (D_k), and policy priors (E_k). Payoff enters as an observation modality rather than an external reward cache, so the agent's preferences are part of the same inference problem as its partner beliefs. State and policy inference use the official `pymdp` discrete active-inference implementation [12, 6].

2.3 Generative model specification

For each partner k, the focal agent maintains a discrete generative model with three latent factors: partner type $s_k^{\text{type}} \in \{\text{cooperator}, \text{reciprocator}, \text{exploiter}, \text{random}\}$, partner stance $s_k^{\text{stance}} \in \{\text{trusting}, \text{neutral}, \text{hostile}\}$, and a deterministic own-action bookkeeping factor s^{own}. The own-action factor is not a psychological hidden variable; it makes the focal agent's executed action available to the payoff likelihood in a standard factorized POMDP representation. Observations are partner action $o_k^{\text{act}} \in \{\text{cooperate}, \text{defect}\}$ and focal payoff $o_k^{\text{pay}} \in \{-1, 1, 3, 5\}$.

The partner-action likelihood A_k^{act} is defined by type- and stance-conditioned cooperation probabilities. This table operationalizes predictive reliability: both reliably cooperative and reliably hostile partners can be predictable, even though their payoff consequences differ. Payoff enters as an observation modality $P(o_k^{\text{pay}} \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}})$, with partner action marginalized through the cooperation likelihood. Thus payoff is deterministic in the environment but represented as a likelihood in the focal agent's generative model. Stance transitions are action-dependent: cooperation tends to build trust, defection tends to damage trust, and recovery from hostility is slower than collapse. Full $A/B/C/D/E$ matrices, cooperation probabilities, stance-transition matrices, priors, and payoff construction are provided in Appendix A.

2.4 Partner-local affective precision

Hesp et al. [13] treat policy precision as an inferred quantity rather than a fixed hyperparameter. The agent maintains a higher-order belief β over inverse policy precision, representing uncertainty about subjective model fitness: how reliably the generative model is supporting confident action. An affective charge signal ϕ drives updates to β, shifting the agent toward higher confidence when the model performs better than a prior baseline and toward lower confidence when it performs worse.

We extend this to the per-partner case by giving each partner its own β_k estimate, updated from that partner's behavioral evidence alone. After interacting with partner k, the agent computes surprisal from the predicted probability of the observed partner response:

$$\epsilon_k = -\log P\big(o_k^{\text{act}} = \hat{o}_k^{\text{act}} \bigm| q(s_k^{\text{type}}, s_k^{\text{stance}})\big) . \tag{1}$$

This is computed from partner *action* rather than payoff because action is the most direct test of type-by-stance model fit: a reliably defecting partner produces no action surprise even though the payoff is poor. We set the neutral fitness baseline at $\sigma_0^2 = (-\log 0.5)^2$, so a fifty-fifty binary prediction carries zero affective charge. Affective charge is then:

$$\phi_k = \alpha\big(\sigma_0^2 - \epsilon_k^2\big) , \tag{2}$$

where α is a gain parameter. Positive ϕ_k means partner k's model predicted better than the baseline; negative ϕ_k means it predicted worse. The categorical posterior $q(\beta_k)$ is updated by combining a slowly evolving persistence prior with a charge-based pseudo-likelihood that shifts mass toward low β after accurate prediction and toward high β after surprise; the full update rule is given in Appendix B.

The agent then uses the posterior mean to set policy precision:

$$\gamma_k = \gamma_{\text{base}} \,/\, \mathbb{E}_{q(\beta_k)}[\beta_k]. \tag{3}$$

Here, β_k represents inverse precision or model-fitness uncertainty: higher β_k means the model has been less reliable, producing a broader, less committed policy distribution; it is the inverse relationship that drives action sharpening. The mechanism thus translates accumulated evidence about predictive reliability into action confidence—independently for each social relationship.

The β_k posterior is maintained as an external auxiliary tracker rather than a proper hidden state in the generative model (discussed in Section 4).

Cross-partner policy selection. In partner-choice regimes, each candidate policy branch inherits the precision of the partner it targets. Precision is applied to centered branch logits so that γ_k sharpens or flattens policies within a partner branch without changing that branch's mean score in cross-partner comparison. Details are provided in Appendix B.

Partner implementation. Reported experiments use one focal active-inference agent with per-partner generative models. Partners are parameterized environment-side policies, not reciprocal active-inference agents: they maintain latent type and stance, sample actions from type-by-stance cooperation probabilities, and undergo action-dependent stance transitions, but they do not run variational inference or maintain their own affective precision estimates. This isolates the focal agent's precision-tracking mechanism before adding reciprocal belief and policy updating on the partner side. Full implementation details are provided in Appendix A.

Simulation scale. Unless otherwise stated, simulations use 200-round episodes. Phenotype and mixed-volatility profiles use 20 seeds per condition, while central behavioral confirmations use 30 seeds per condition. The abrupt-betrayal condition uses 120 rounds because the switch is scheduled explicitly and the main readouts are post-switch behavior. Full condition definitions, schedules, metrics, and seed counts are provided in Appendix C.

Code availability. All simulations, analysis scripts, and experiment configurations are available at `https://github.com/har5h1l/affect_aif`.

3 Results

Simulations test six linked predictions about relationship-specific confidence: whether precision tracks predictive reliability rather than value, whether locality matters, whether behavioral effects arise through action sharpening rather than improved beliefs, how precision shapes partner engagement, how abrupt change exposes temporal dependencies, and how precision parameters define social-trust profiles.

3.1 Precision tracks predictability, not partner value

The affective charge formula uses action surprisal rather than payoff, making β_k sensitive to partner behavioral consistency regardless of whether that consistency is rewarding. A reliably defecting partner produces near-zero ϵ_k once the agent has inferred its type, yielding high γ_k despite poor payoff. An erratic partner generates high ϵ_k and low γ_k regardless of its average payoff. The precision signal is therefore designed to track the quality of the agent's predictive model, not the desirability of the partner's behavior.

The partner-local implementation preserves this signal most cleanly. In the focal-switch locality probe, the surprise-over-reward correlation gap is sharp for local β_k: $|\mathrm{corr}(\mathrm{precision}, \mathrm{surprise})| = 0.943$ versus $|\mathrm{corr}(\mathrm{precision}, \mathrm{payoff})| = 0.110$ across partner-seed units. A single shared β tracker is considerably weaker on the same readout (0.149 versus 0.043), showing that pooling evidence across partners dilutes the reliability signal. Aggregate payoff does not distinguish partner-local β_k from shared β in this low-interference probe (partner-local: 946.8; shared: 976.2; no-affect: 950.7), motivating the shared-β ablation below, where evidence from stable and volatile partners competes within a single social portfolio. Figure 1 visualizes this dissociation for local versus shared precision tracking.

3.2 Shared precision tests relational specificity

The locality probe above shows that pooling evidence across partners weakens the reliability readout, but it does not by itself establish a behavioral cost because cross-partner interference is low in that setting. The stronger test compares partner-local β_k against a shared β tracker in regimes where stable, hostile,

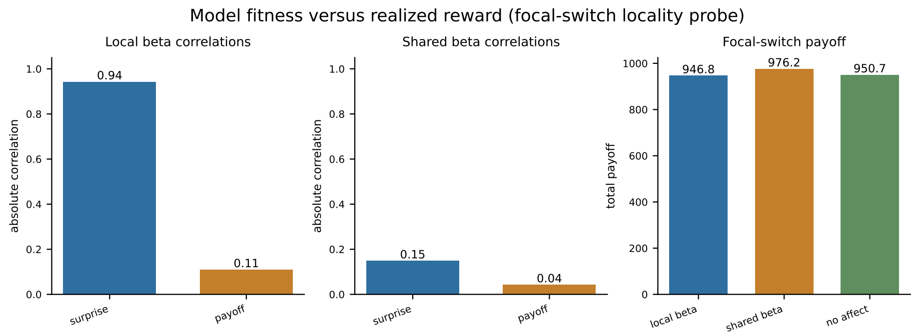

Fig. 1: Partner-local precision tracks action surprisal more strongly than realized payoff. The panel contrasts absolute associations between mean partner precision and surprise versus payoff for local β_k and a shared β tracker. Local tracking preserves the surprise-over-reward dissociation; pooling evidence across partners attenuates relational specificity.

and changing partners coexist. If the proposed mechanism requires relational specificity, the shared tracker should blur evidence across the social portfolio: confidence lost toward volatile partners should spill over onto stable partners, while confidence earned from stable partners should slow adaptation to changing ones.

[*Shared-β ablation results pending: report partner-local versus shared precision on payoff, entropy, reliability-specificity, partner allocation, and false-positive suppression in the mixed-volatility or focal-switch regime.*]

3.3 Behavioral gains arise through action sharpening

A tracked-only ablation updates $q(\beta_k)$ normally but fixes $\gamma_k = \gamma_{\text{base}}$, allowing the precision estimate to be analyzed without letting it affect action selection. In the open graded decision regime (three-seed exploratory diagnostic), full partner-local precision lowers policy entropy relative to both baselines (8.6 versus 8.8 for no-affect and tracked-only), while total payoff is flat-to-slightly lower (1851.3 versus 1864.2 for both baselines). Because tracked-only remains near the no-affect baseline while full precision modulation differs from both on entropy, the behavioral effect is localized to the $\beta_k \to \gamma_k$ deployment pathway rather than improved partner-state inference. Figure 2 summarizes the three-way contrast across affect, no-affect, and tracked-only variants.

3.4 Partner engagement reorganizes around reliability

In partner-choice regimes the precision signal shapes not only how the agent behaves toward a chosen partner but which partners it selects in the first place.

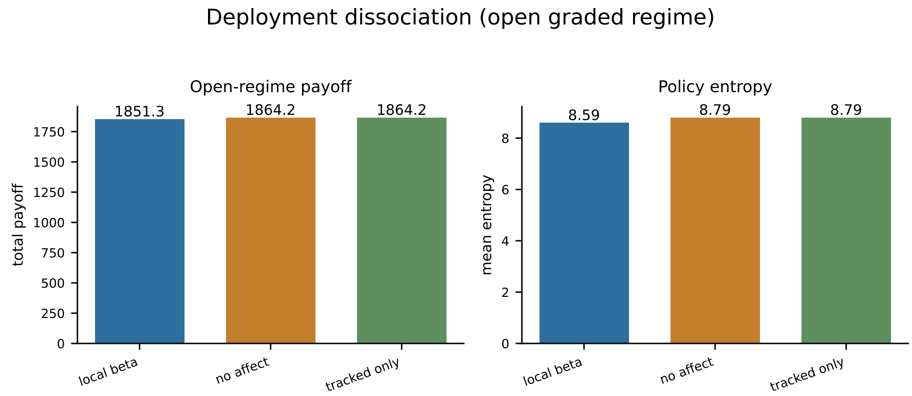

Fig. 2: Open graded regime (three-seed exploratory diagnostic): partner-local affective precision lowers policy entropy relative to no-affect and tracked-only controls without a corresponding payoff advantage. Tracked-only updates β_k but fixes γ_k; its deployment readouts match the no-affect baseline, isolating the entropy effect to the $\beta_k \to \gamma_k$ action-confidence pathway.

Policy entropy under precision modulation is 4.66 versus 4.81 without it (three-seed exploratory diagnostic), and engagement shifts in the expected direction, with allocation moving toward more predictable partners and away from less predictable ones. Aggregate payoff is flat at this scale (377.3 versus 385.3), consistent with selection reorganization occurring before cumulative payoff differences have time to accumulate.

[*Partner-choice allocation rates pending: report selection share by partner under affect and no-affect, and whether payoff remains flat at this scale.*]

3.5 Abrupt betrayal reveals the mechanism's temporal structure

When a previously cooperative partner switches stance mid-episode, the agent enters the post-switch period with β_k reflecting a well-validated model that no longer applies. In the abrupt betrayal condition, partner-local affective precision improves total payoff relative to the no-affect control (1322.3 versus 1225.0), driven by sharper reallocation toward reliable alternatives before the no-affect agent has accumulated sufficient evidence to shift engagement. Policy entropy is lower under precision modulation (7.47 versus 8.68), confirming that the action-confidence pathway is active.

The accuracy pattern is theoretically important. The affective agent achieves higher payoff while maintaining lower joint accuracy (0.319 versus 0.425): the gain does not arise from better partner-type inference but from acting more decisively on accumulated reliability evidence across the social portfolio. Figure 3 shows this payoff–accuracy dissociation under abrupt betrayal. The abrupt-change condition is therefore best interpreted as a temporal-dependency result rather than a

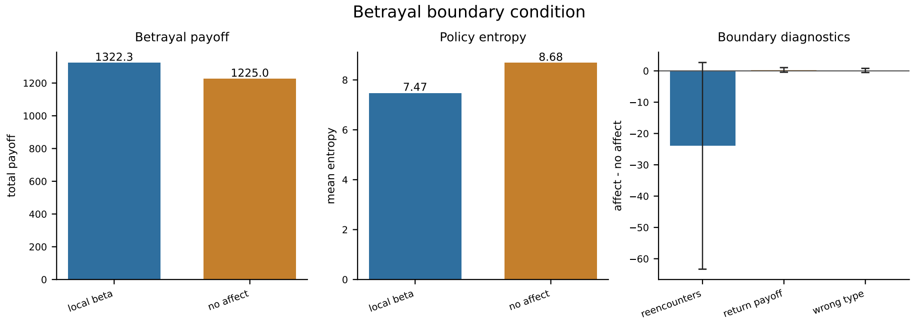

Fig. 3: Abrupt betrayal in the partner-choice regime. Partner-local affect achieves higher total payoff than the no-affect control while the main-text accuracy readout shows lower joint partner-type accuracy, consistent with adaptive reallocation across the social portfolio rather than superior inference about the changed partner. Policy entropy is lower under precision modulation (7.47 versus 8.68).

simple success-or-failure test of affective precision. Accumulated confidence helps when it can be redirected toward reliable alternatives quickly enough, and it becomes risky when confidence remains attached to a stale partner model with no productive alternative route for action.

3.6 Precision dynamics as individual differences in social trust calibration

The parameters governing β_k dynamics define a controlled phenotype program. Variation in precision-gain α and the prior over model fitness produces qualitatively distinct behavioral profiles across social contexts. These profiles are computational analogues of empirically characterized individual differences in social trust calibration [3, 1, 5], not clinical classifications.

Phenotype-inspired computational perturbations (Appendix D) show that flattened precision dynamics can collapse action commitment without improving payoff: the alexithymia-like variant compresses β_k range to 0.24 and raises entropy to 8.9, near the no-affect and tracked-only baselines, whereas expanded-dynamics variants widen β_k movement without monotonic payoff gains. This motivates parameter sweeps over gain and prior structure rather than a single high-versus-low trust axis.

Precision-gain axis. Sweeping α across $\{0.05, 0.1, 0.3, 0.5, 1.0, 2.0, 4.0, 8.0\}$ shows that β_k movement range increases monotonically with α in the betrayal regime. Low-α agents ($\alpha \leq 0.1$) show near-zero exploitation rates and Gini coefficients of 0.423 and 0.397, producing diffuse, slowly-updating engagement patterns. High-α agents ($\alpha \geq 4.0$) show Gini coefficients of 0.366 and 0.408 with recovery times of 28.8 and 13.0 rounds respectively—faster detection of behavioral change in some runs, but higher variance in recovery outcomes. Payoff does not rank α

values monotonically across either regime (betrayal: range 1914.6–1986.2; open graded: 1906.3–1987.7), indicating that the behavioral consequence of precision-gain depends on task context rather than gain magnitude alone. Extended alpha-sweep diagnostics are in Appendix D.

Phenotype quadrants. Crossing prior belief (naive versus cautious) with α (low versus high) defines four behavioral profiles tested in the betrayal environment (20 seeds). Low-α phenotypes (naive-stubborn and avoidant-rigid) show substantially smaller β_k range (0.32–0.36) than high-α phenotypes (0.89–1.08), confirming that gain controls the magnitude of precision dynamics. Naive-stubborn shows the lowest Gini (0.304), reflecting maximally diffuse partner selection consistent with a slowly-updating confidence profile. Avoidant-rigid shows the lowest payoff (1889), as the cautious prior prevents confidence from building even toward reliably cooperative partners. The anxious-reactive profile shows the highest post-betrayal commitment errors toward the switched partner (post-betrayal P0 selection: 0.319; high investment rate: 0.121), reflecting rapid confidence accumulation that becomes a liability when a partner's behavior changes. The default agent outperforms all phenotype variants on payoff (1991), occupying a calibrated region of the gain-prior space. Quadrant figures and supporting diagnostics are in Appendix D.

Forgiveness and trust repair. The forgiveness/trust-repair condition is designed to separate willingness to reengage from restoration of confidence; final metrics are reported in Appendix D.

Mixed-volatility discrimination. The most ecologically realistic test places four partners with heterogeneous volatility in the same episode: a stationary cooperator (P0), a stationary exploiter (P1), an abrupt-switch partner (P2), and a gradual-drift partner (P3). Under partner choice, the agent must maintain stable engagement with P0 while updating away from P2 and P3 after their behavioral changes.

Table 2: Mixed-volatility metrics by precision profile (20 seeds).

Variant	Payoff	Discrimination	P0 false positive	β range
default	1976	0.121	0.422	0.933
high-α	2054	−0.089	0.484	1.049
low-α	1927	0.213	0.359	0.182
no-affect	1934	–	0.273	–

The false-positive rate—unwarranted confidence reduction toward the stationary P0—increases with α: high-α (0.484), default (0.422), low-α (0.359), no-affect (0.273). The discrimination index does not simply improve with gain: low-α gives

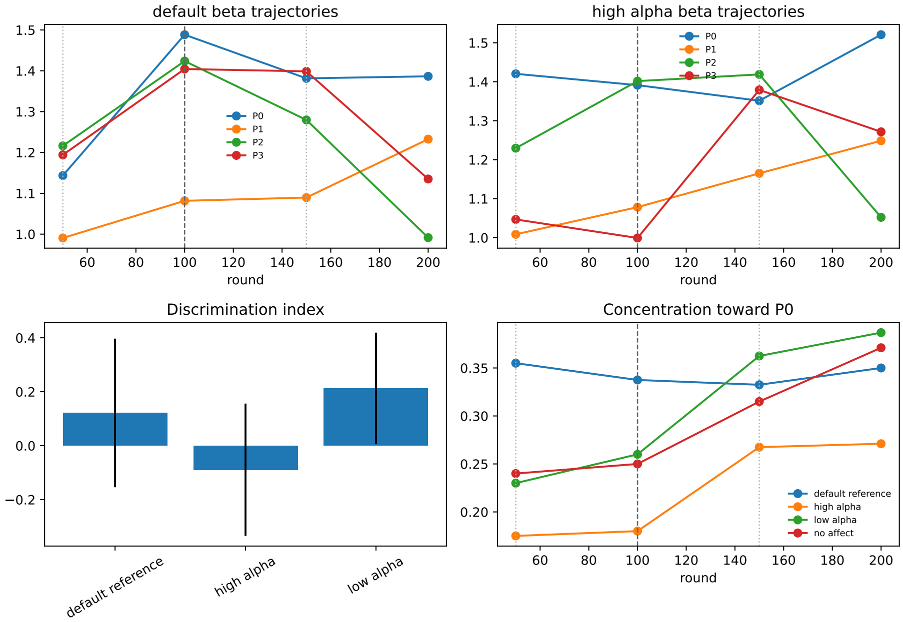

Fig. 4: Mixed-volatility partner-choice environment (20 seeds). Four partners differ in volatility: stationary cooperator (P0), stationary exploiter (P1), abrupt switch (P2), and gradual drift (P3). Higher precision-gain α can raise payoff while worsening discrimination and false-positive suppression toward P0; profiles are computational analogues, not clinical classifications.

the strongest discrimination score (0.213), while high-α is negative (-0.089) despite achieving the highest payoff. Figure 4 shows why the result should not be read as a monotonic gain advantage. The mixed-volatility result therefore supports a narrower conclusion than the original sensitivity-specificity prediction: high gain amplifies reactivity and false positives toward stable partners, but the payoff consequences depend on how that reactivity interacts with partner allocation.

Together, the phenotype experiments characterize affective precision dynamics not as a single behavioral tendency but as a family of profiles determined by gain and prior initial conditions. The key result is not a monotonic payoff ranking, but a dissociation between confidence dynamics, partner allocation, and payoff across social environments. Extended phenotype tables, alpha-sweep diagnostics, and forgiveness/trust-repair analyses are provided in Appendix D.

4 Discussion

Affect as social metacognition. Partner-local affective precision is best understood as social metacognition: an estimate not of what a partner will do, but of how well the focal agent's current model of that partner justifies confident action. Unlike theory-of-mind inference, which represents another agent's beliefs, knowledge, or intentions [10, 17], this signal evaluates the reliability of the focal agent's own partner model. It is therefore metacognitive rather than mentalistic.

The two mechanisms are complementary. An agent with sophisticated ToM inference could still be poorly calibrated about when to act confidently on those inferences, particularly if the partner is reliable in some contexts and unreliable in others. Partner-local affective precision addresses exactly this gap: it allows action confidence to track the recent track record of each partner model, independently of the model's representational content. This complementarity is closest to ToM active-inference accounts of social cognition [17], while factorized active-inference models provide a formal route for separating multiple latent components of strategic interaction [18]. The combination would permit an agent to both model partner intentions and maintain a running estimate of how much those models currently justify decisive action—a richer form of social metacognition that may be necessary for robust performance in complex multi-partner environments.

Individual differences and the computational basis of social trust variation. The phenotype program in Section 3.6 does not claim that the four simulated profiles are validated clinical categories. Its contribution is more specific: it shows how two parameters, precision gain α and the prior over model fitness, can vary the *speed*, *amplitude*, and *asymmetry* of social confidence revision. This is the distinction a scalar trust account misses. Two agents can have similar average trust while differing sharply in whether they update quickly after negative evidence, recover after renewed cooperation, or overreact to noise from otherwise stable partners.

This framing connects the simulations to empirical literatures on social learning, social anxiety, psychosis, and attachment without reducing those constructs to model labels [3, 1, 5]. These paradigms matter because they separate dimensions that scalar trust models collapse: reengagement versus restored confidence, sensitivity to change versus false positives, and payoff improvement versus accurate discrimination.

Betrayal reveals a temporal dependency in confidence deployment. The betrayal result makes this distinction concrete: local affect improves payoff while lowering joint accuracy, showing that the mechanism can improve action allocation without improving inference about the changed partner. Accumulated β_k evidence from the pre-betrayal cooperative phase changes action commitment across the social portfolio, allowing engagement to shift toward reliable alternatives even while the local model of the changed partner remains imperfect. The implication is structural. A confidence mechanism built from evidence accumulation has a temporal dependency: its benefit under abrupt change depends on whether accumulated confidence can be productively redirected, whether alternative partners exist, and whether reallocation occurs faster than stale confidence can keep action attached to the changed partner. This suggests a natural extension: a volatility or change-detection channel that discounts accumulated confidence when prediction-error structure changes. This is related to the problem addressed by hierarchical volatility models [15, 3], but it appears here at the level of action confidence rather than learning rate.

Limitations. Three design features bound the current claims. First, β_k is an external auxiliary tracker; in the full Hesp et al. hierarchy it is a proper inferred hidden state, allowing the agent to form prior expectations over its own future affective states and to plan around anticipated changes in partner reliability. Making β_k a variational variable would enable richer prospective regulation of action confidence. Second, partners are parameterized policies rather than themselves being active-inference agents. This isolates the focal mechanism, but it means the present simulations do not address the reciprocal case in which both agents update their beliefs, confidence, and policies simultaneously. Third, the phenotype taxonomy in Section 3.6 is currently predictive rather than validated: the experiments test whether different α and prior values produce qualitatively distinct behavioral profiles, and the manuscript connects those targets to human individual differences in social trust. A stronger claim—that individual differences in human social trust calibration arise specifically from variation in an affective precision mechanism with this structure—would require fitting the model to human behavioral data and establishing that parameter estimates predict theoretically relevant outcomes. The phenotype predictions are offered as a scaffold for such empirical work, not as an established correspondence.

Future directions. The most important near-term extension is empirical: fitting the affective precision model to human behavioral data from iterated trust game paradigms with individual-difference measures. The phenotype analyses in Section 3.6 generate quantitative predictions about exploitation sensitivity, betrayal recovery time, trust-revision asymmetry, and mixed-volatility false positives that are measurable from behavioral time series without neuroimaging. The α and prior parameters could be estimated per participant and tested against personality measures, attachment style, and clinical scores to assess whether the computational phenotypes correspond to real psychological variation. A second extension is ecological and reciprocal: replacing environment-side partner policies with active-inference partners that maintain their own beliefs, precision estimates, and policies, then embedding these agents in settings with communication, reputation, and coalition structure. Combining this signal with ToM-style belief inference [17] would allow future work to study whether metacognitive action-confidence signals and mentalistic partner-state models are functionally coupled or dissociable in social decision-making.

5 Conclusion

Predicting what a partner will do and deciding how confidently to act on that prediction are separable problems. Partner-local affective precision addresses the second. By giving each social relationship its own model-fitness estimate, the mechanism allows action confidence to be calibrated independently across a social portfolio—more decisive toward partners whose behavior the model predicts well, more cautious toward those who remain uncertain. Current evidence supports reliability tracking, metacognitive rather than epistemic deployment, and a temporal dependency on social portfolio structure under change. The parameters governing these dynamics—precision gain and prior over model fitness—define a family of behavioral phenotype targets for individual-difference research, positioning the framework as a mechanistically grounded and empirically tractable account of social affect.

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Appendix

A Full POMDP Specification

Scope. This appendix provides supporting technical detail for the corresponding main-text sections. Essential claims and results remain in the main paper.

A.1 Hidden state factors and observations

The focal agent's generative model for each partner k maintains three latent factors:

$$\begin{aligned} s_k^{\text{type}} &\in \{\text{cooperator, reciprocator,} \\ &\qquad\quad \text{exploiter, random}\} && \text{(latent behavioral type)} && (4) \\ s_k^{\text{stance}} &\in \{\text{trusting, neutral, hostile}\} && \text{(latent stance)} && (5) \\ s^{\text{own}} &\in \mathcal{A} && \text{(agent's executed social action)} && (6) \end{aligned}$$

The own-action state is *deterministic bookkeeping*: the agent always knows its own action, and this factor makes that knowledge available to the observation model. It is not a psychological hidden variable. In the binary task, the joint hidden state per partner spans $4 \times 3 \times 2 = 24$ possibilities; graded regimes replace the binary action factor with the configured investment levels.

Observations are partner action and payoff:

$$\begin{aligned} o_k^{\text{act}} &\in \{\text{cooperate, defect}\} && \text{(partner response)} && (7) \\ o_k^{\text{pay}} &\in \{-1, 1, 3, 5\} && \text{(focal agent's realized payoff)} && (8) \end{aligned}$$

A.2 Partner-action likelihood

The core of the generative model is the *partner cooperation likelihood table*, which defines the probability the partner cooperates given type and stance:

Table 3: Cooperation likelihood $P(\text{cooperate} \mid s^{\text{type}}, s^{\text{stance}})$.

Partner type	Trusting	Neutral	Hostile
Cooperator	0.95	0.80	0.55
Reciprocator	0.90	0.70	0.30
Exploiter	0.70	0.35	0.10
Random	0.60	0.50	0.35

This table is the foundation of the likelihood modality A_k^{act}. Entries define

$$P\left(o_k^{\text{act}} = \text{cooperate} \mid s_k^{\text{type}}, s_k^{\text{stance}}\right), \tag{9}$$

with $P(o_k^{\text{act}} = \text{defect} \mid \cdot) = 1 - P(o_k^{\text{act}} = \text{cooperate} \mid \cdot)$. The modality is uniform over s^{own}: partner action depends only on inferred type and stance. This operationalizes what the agent considers a predictable partner—a cooperative reciprocator in a trusting stance has high predicted cooperation; an exploiter in a hostile stance has low predicted cooperation.

A.3 Payoff likelihood

Payoff enters as a second observation modality $A_k^{\text{pay}} = P(o_k^{\text{pay}} \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}})$ rather than as external reward. The environment payoff table is deterministic (Table 1 in the main text); in the generative model, partner action is marginalized through the cooperation likelihood:

$$P\left(o_k^{\text{pay}} = v \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}}\right) = \sum_{a^{\text{partner}} \in \{\text{C},\text{D}\}} \mathbb{I}\left[\pi(s^{\text{own}}, a^{\text{partner}}) = v\right] P\left(o_k^{\text{act}} = a^{\text{partner}} \mid s_k^{\text{type}}, s_k^{\text{stance}}\right), \tag{10}$$

where $\pi(\cdot,\cdot)$ maps joint actions to focal payoffs $\{-1, 1, 3, 5\}$ and $P(o_k^{\text{act}} \mid \cdot)$ comes from Eq. (9). This preserves the Prisoner's Dilemma incentive structure at horizon 1 while allowing trust-building dynamics to emerge over longer horizons.

A.4 Transition dynamics

Type drift (B_k^{type}). Partner type is uncontrollable and drifts slowly:

$$P(s_k^{\text{type}\prime} = j \mid s_k^{\text{type}} = i) = \begin{cases} 1 - p_{\text{switch}} & j = i, \\ p_{\text{switch}}/(K_{\text{type}} - 1) & j \neq i, \end{cases} \tag{11}$$

with $K_{\text{type}} = 4$ and default $p_{\text{switch}} = 0.05$. Scheduled betrayal and mixed-volatility scenarios set $p_{\text{switch}} = 0$.

Stance dynamics (B_k^{stance}). Stance transitions are action-dependent on the focal agent's social action a:

$$P\left(s_k^{\text{stance}\prime} \mid s_k^{\text{stance}}, a\right), \tag{12}$$

as specified below. Cooperation tends to build trust; defection damages trust; recovery from hostility is slower than collapse.

Table 4: Stance transition dynamics $P(s'^{\text{stance}} \mid s^{\text{stance}}, a)$ conditional on focal agent action.

Agent cooperates			
To \ From	Trusting	Neutral	Hostile
Trusting	0.90	0.30	0.05
Neutral	0.10	0.60	0.35
Hostile	0.00	0.10	0.60
Agent defects			
To \ From	Trusting	Neutral	Hostile
Trusting	0.10	0.05	0.02
Neutral	0.50	0.35	0.18
Hostile	0.40	0.60	0.80

These dynamics instantiate the asymmetry of trust: cooperation gradually builds trust (neutral $\to$ trusting in ~ 3–4 steps), while defection rapidly damages trust (trusting $\to$ hostile in ~ 2 steps); recovery from hostility is slow (~ 4–5 steps to neutral).

Own-action bookkeeping (B^{own}). The own-action factor updates deterministically from the executed social action:

$$P(s^{\text{own}\prime} = a' \mid s^{\text{own}}, a) = \mathbb{I}[a' = a]. \tag{13}$$

This factor is not a psychological hidden variable; it makes the agent's executed action available to the payoff likelihood in standard factorized form.

A.5 Preferences, priors, and policy priors

Observation preferences (C_k). There is no direct preference over partner actions:

$$C_k^{\text{act}} = [0,\ 0] \quad \text{over } \{\text{cooperate}, \text{defect}\}. \tag{14}$$

Payoff preferences ascend over $\{-1, 1, 3, 5\}$:

$$C_k^{\text{pay}} = \text{log\_softmax}\left(\frac{[-1, 1, 3, 5]}{\tau}\right), \tag{15}$$

where τ is a temperature parameter controlling preference sharpness. Instrumental motivation therefore enters through payoff observations and multi-step planning rather than direct action preferences.

Initial-state priors (D_k). Type, stance, and own-action priors are

$$D_k^{\text{type}} = \left[\tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}\right], \tag{16}$$

$$D_k^{\text{stance}} = [0.2,\ 0.6,\ 0.2], \tag{17}$$

$$D^{\text{own}} = [0.5,\ 0.5] \quad \text{(binary task; uniform over graded levels in graded regimes)}. \tag{18}$$

Policy priors (E_k). Policy priors are uniform over admissible action sequences at the configured planning horizon (no habitual bias toward cooperation or defection). For horizon 1 in the binary task, $E_k = [0.5, 0.5]$ over $\{\text{cooperate}, \text{defect}\}$; longer horizons enumerate all length-τ sequences with equal prior mass.

Partner selection across k is handled outside the per-partner POMDP by evaluating policy branches for each partner with precision-adjusted logits before executing the chosen action (Appendix B).

A.6 Partner implementation

Reported experiments use one focal active-inference agent with per-partner generative models as described above. Partners are parameterized environment-side policies, not reciprocal active-inference agents. Each partner maintains a latent type and stance, follows the type-by-stance cooperation probabilities in Table 3, and undergoes action-dependent stance transitions in Table 4. Partners do not run variational inference, maintain their own policy precision γ_k, or update affective estimates β_k. This design isolates the focal agent's precision-tracking mechanism before adding reciprocal belief and policy updating on the partner side.

B Affective Precision Update and Policy Selection

Scope. This appendix provides supporting technical detail for the corresponding main-text sections. Essential claims and results remain in the main paper.

B.1 Categorical beta support

Each partner maintains a categorical posterior $q(\beta_k)$ over discrete levels $\{0.5, 0.67, 1.0, 1.5, 2.0\}$, updated each round from affective charge ϕ_k. The posterior is categorical rather than continuous; this keeps the inverse-β convention identifiable while preserving a compact implementation of the precision dynamics.

B.2 Persistence prior

The update proceeds in three steps. First, a persistence prior smooths the previous posterior via a tridiagonal transition matrix:

$$p(\beta_k) \leftarrow T \cdot q_{\text{prev}}(\beta_k), \tag{19}$$

where T is tridiagonal with persistence parameter 0.8 and reflecting boundaries, encoding the prior expectation that precision evolves slowly.

B.3 Charge-based pseudo-likelihood

Second, a pseudo-likelihood is constructed directly from affective charge, mapping positive charge to high precision (low β) and negative charge to low precision (high β):

$$\log \ell(\beta_\ell) = \phi_k \cdot \frac{1}{\beta_\ell}, \tag{20}$$

where β_ℓ ranges over the five support points. This is a hand-designed update rule (not derived from the POMDP generative model) chosen to satisfy monotonicity: strong positive charge reinforces low β, strong negative charge reinforces high β, and the effect scales with effective precision at each level.

B.4 Posterior update and policy precision

Third, the posterior is normalized:

$$q(\beta_k) \propto \exp\left(\log \ell(\beta_\ell)\right) \cdot p(\beta_k). \tag{21}$$

The agent then uses the posterior mean to set policy precision:

$$\gamma_k = \gamma_{\text{base}} \,/\, \mathbb{E}_{q(\beta_k)}[\beta_k]. \tag{22}$$

The β_k posterior is maintained as an external auxiliary tracker rather than a proper hidden state in the generative model, which simplifies implementation at the cost of losing the agent's ability to form prior expectations over its own future affective states (discussed in Section 4).

B.5 Centered cross-partner policy logits

In partner-choice regimes, precision must sharpen policies within a partner branch without creating artifacts when branches are compared across partners. Let $u_{k,\pi}$ denote the pre-precision policy logit for policy π toward partner k, and let $\bar{u}_k$ be that partner branch's mean logit. Precision is applied to centered logits:

$$\tilde{u}_{k,\pi} = \bar{u}_k + \gamma_k \left(u_{k,\pi} - \bar{u}_k\right). \tag{23}$$

The agent then samples or selects policies from the combined set of $\tilde{u}_{k,\pi}$ values. Centering preserves the mean score of each partner branch while allowing γ_k to sharpen or flatten the within-partner policy distribution. This prevents low-precision scaling from moving a uniformly poor branch toward zero and making it spuriously attractive in cross-partner comparison.

B.6 Ablation variants

Reported comparisons use four affect-runtime variants. The *no-affect* control disables the beta tracker and fixes policy precision at γ_{base}. The *tracked-only* ablation updates $q(\beta_k)$ normally but fixes $\gamma_k = \gamma_{\text{base}}$, allowing the precision estimate to be analyzed without letting it affect action selection. The default *partner-local precision* variant applies the full update and sets γ_k from each partner's posterior mean. The *shared-*β ablation maintains a single pooled beta state updated from partner evidence while keeping partner-local POMDP beliefs intact; all partner branches then inherit precision from the shared expected β.

C Simulation Protocols and Metrics

Scope. This appendix provides supporting technical detail for the corresponding main-text sections. Essential claims and results remain in the main paper.

C.1 Simulation conditions

All reported experiments use one focal active-inference agent with four partners ($K = 4$), per-partner generative models, and environment-side partner policies (Appendix A). Conditions differ by payoff regime, assignment mode, and scheduled partner changes.

Binary random-assignment baseline. In the binary trust game (Table 1 in the main text), partners are assigned randomly each round and the agent chooses cooperate or defect. Stochastic partner-type drift uses $p_{\text{switch}} = 0.05$ unless disabled for scheduled-switch scenarios.

Open graded decision regime. The graded regime replaces the binary action with a discretized investment level while keeping the same latent type–stance structure. This is the primary deployment-dissociation setting (Section 3.3): affect, no-affect, and tracked-only variants are compared on policy entropy and cumulative payoff.

Partner-choice regime. The agent first selects a partner, then selects an action toward that partner each round. Partner selection is where engagement reorganization and social-allocation readouts are defined (Section 3.4).

Focal-switch locality probe. A partner-choice graded environment with type volatility disabled ($p_{\text{switch}} = 0$) and a single scheduled stance switch on partner P0 tests whether local versus shared β tracking preserves the surprise-over-reward dissociation (Section 3.1).

Abrupt betrayal. A partner-choice graded environment with scheduled stance betrayal on a previously cooperative partner tests post-switch reallocation, entropy, payoff, and joint accuracy (Section 3.5).

Precision-dynamics perturbations. Clinical-like computational variants (alexithymia-like, borderline-like, depression-like) perturb β_k amplitude and stability in the graded decision regime without claiming clinical validity (Appendix D).

Phenotype program (Exp A–D). Parameter sweeps and factorial designs vary precision gain α and prior over model fitness to define computational trust-calibration profiles in open graded, betrayal, partner-choice, forgiveness, and mixed-volatility environments (Section 3.6).

C.2 Seed counts and episode lengths

Unless otherwise stated in the main text, episode length is 200 rounds. The abrupt-betrayal condition uses 120 rounds because post-switch readouts dominate interpretation. Phenotype and mixed-volatility tables use 20 seeds per condition. Central behavioral confirmations target 30 seeds per condition.

Exploratory diagnostic runs for the H0–H6 spine used three seeds per variant for exploratory validation of the surprisal-based precision tracker and centered cross-partner policy selection; those numbers remain underpowered and are not treated as confirmation-scale evidence. Current manuscript headline values drawn from exploratory diagnostic runs are flagged for replacement where 30-seed confirmation is pending (Section 3).

Representative configured scales:

Table 5: Representative episode lengths and replication counts by experiment family.

Experiment family	Setting	Rounds	
H0/H2 open graded deployment	graded, random or open choice	200	30 (confirm.) / 3 (explo
H1 model fitness	graded / reliability probes	200	30 (confirm.) / 3 (explo
H3 focal-switch locality	partner choice, single switch	100	30 (confirm.) / 3 (explo
H4 partner choice	partner choice	200	30 (confirm.) / 3 (explo
H5 abrupt betrayal	partner choice, scheduled betrayal	120	30 (confirm.) / 3 (explo
H6 clinical dynamics	graded perturbations	200	30 (confirm.) / 3 (explo
Exp A–D phenotypes	sweeps / factorial / forgiveness / mixed volatility	200	

C.3 Betrayal schedule

The canonical abrupt-betrayal protocol (H5 `betrayal_choice`) uses graded payoffs, agent-chosen partners, four partners with fixed initial types and stances, and no stochastic type switching ($p_{\text{switch}} = 0$). Partner P0 begins as a cooperator in a trusting stance. At round 31, P0 undergoes a scheduled stance switch to hostile while other partners retain their initial configurations. Primary readouts are post-switch cumulative payoff, policy entropy, partner reallocation, and joint partner-type accuracy over the remaining episode.

The phenotype betrayal environment used in Exp A and Exp B applies the same single-switch structure within 200-round episodes for recovery-time and quadrant metrics. Exp C (forgiveness) extends this schedule: rounds 1–80 cooperative baseline, rounds 81–120 betrayal on P0, rounds 121–200 reversion to cooperative/trusting on P0.

C.4 Mixed-volatility schedule

The mixed-volatility environment (Exp D; Section 3.6) places four partners with heterogeneous volatility in the same partner-choice episode over 200 rounds:

- **P0:** stationary cooperator (no scheduled change).
- **P1:** stationary exploiter (no scheduled change).
- **P2:** abrupt shift—cooperative/trusting pre-switch, hostile/exploiter from the scheduled switch onward (round 101 in the implemented scenario).
- **P3:** gradual drift—stance and type transitions spread across mid-episode rounds (neutral stance from round 51; hostile/exploiter shift from round 151 in the implemented scenario).

Stochastic type switching is disabled ($p_{\text{switch}} = 0$). Conditions compare default, low-α, high-α, and no-affect variants at 20 seeds each.

C.5 Partner-choice allocation metrics

Partner-choice readouts quantify how precision reshapes *who* the agent engages, not only how it acts toward a fixed partner.

Selection share. Per-partner selection rate is the fraction of rounds on which each partner is chosen. Pending confirmation-scale reporting includes affect versus no-affect selection shares by partner (Section 3.4).

Selection concentration. The Gini coefficient of the partner-selection distribution summarizes how concentrated engagement is across the social portfolio (used in phenotype Exp A and Exp B).

Post-betrayal allocation. Post-switch windows report reallocation away from the switched partner and toward alternatives, including post-betrayal P0 selection and high-investment rates in phenotype analyses.

Policy entropy under choice. Mean q_π entropy in partner-choice regimes captures how sharply the agent commits to partner–action policies (reported comparatively for affect versus no-affect controls).

C.6 Entropy, payoff, and accuracy readouts

Total payoff. Cumulative focal-agent payoff over the episode (or over defined post-switch windows for betrayal analyses).

Policy entropy. Mean q_π entropy (‘q_pi_entropy‘ in logged outputs) measures action-commitment sharpness; lower entropy indicates sharper policy selection under precision modulation.

Joint partner-type accuracy. Mean joint correctness of inferred partner type (‘mean_joint_accuracy‘) summarizes epistemic state inference; it is reported separately from payoff because the betrayal result can show higher payoff with lower accuracy when reallocation succeeds.

β_k **range.** Per-partner or episode-aggregated range of β_k support summarizes precision-dynamics amplitude in phenotype and perturbation tables.

Model-fitness dissociation. Partner-local correlations between mean precision and action surprisal versus realized payoff test whether β_k tracks predictive reliability rather than partner value (Section 3.1).

Mixed-volatility discrimination metrics. The discrimination index is the Pearson correlation between per-partner β_k trajectory and per-partner behavioral stability (higher indicates better separation of stable versus shifting partners). The P0 false-positive rate is the rolling rate at which engagement with the stationary cooperator P0 falls more than 15% below its early baseline despite P0 never changing.

Betrayal timing metrics. Recovery time and related latency readouts measure rounds after a switch until selection or payoff returns toward baseline thresholds (phenotype Exp A/B and betrayal analyses).

C.7 Placeholder results awaiting final runs

The following readouts remain pending confirmation-scale values. Main-text placeholders and TODO comments are retained in Section 3; this list records what still awaits final runs without filling numbers.

- **Shared-β ablation** (Section 3.2): partner-local versus shared precision on payoff, entropy, reliability-specificity, partner allocation, and false-positive suppression in mixed-volatility or focal-switch regimes.
- **Partner-choice allocation rates** (Section 3.4): selection share by partner under affect and no-affect, and whether aggregate payoff remains flat at confirmation scale.
- **Forgiveness / trust-repair experiment** (Section 3.6; Appendix D): reengagement latency, post-return selection, payoff recovery, and whether high-α controls reengagement speed while cautious priors limit the trust-repair ceiling.
- **H1 locality-probe confirmation**: replace focal-switch surprise–payoff correlation and aggregate payoff values with final 30-seed confirmation if they differ from current exploratory diagnostic values.
- **H5 betrayal confirmation**: replace abrupt-betrayal payoff, entropy, and joint-accuracy values with final 30-seed H5 confirmation values when available.

D Extended Phenotype and Robustness Results

Scope. This appendix provides supporting technical detail for the corresponding main-text sections. Essential claims and results remain in the main paper.

D.1 Precision-dynamics perturbations

To connect the basic mechanism to the phenotype program, we perturb the amplitude and stability of β_k dynamics using phenotype-inspired computational variants. These variants are not diagnostic models; they are controlled ways of asking how flattened, expanded, or unstable precision dynamics alter action commitment.

Table 6: Precision-dynamics perturbations in the graded decision regime. Variants are computational profiles, not clinical diagnoses.

Variant	β_k range	Total payoff	Policy entropy
affect	1.32	1851.3	8.6
alexithymia-like	0.24	1840.5	8.9
borderline-like	1.48	1859.0	8.1
depression-like	1.48	1877.7	8.4

Signal flattening produces the clearest behavioral consequence. The alexithymia-like variant compresses β_k range to 0.24 and raises entropy to 8.9, near the no-affect and tracked-only baselines. Expanded-dynamics variants produce wider β_k movement, but payoff does not follow range monotonically. This motivates the parameter sweeps below: social confidence differences are better captured by gain and prior structure than by a single high-versus-low trust axis.

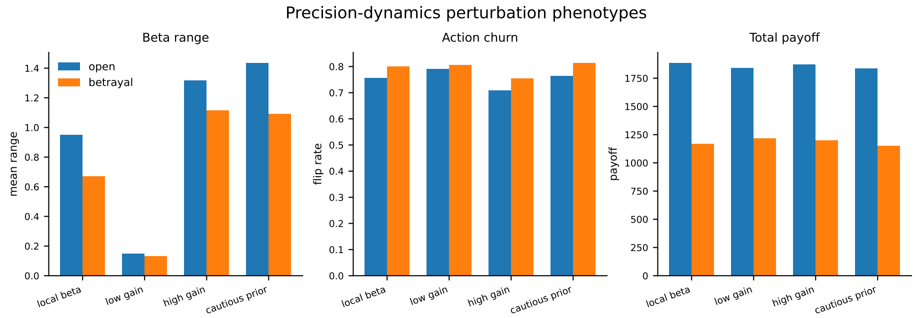

Fig. 5: Precision-dynamics perturbations in the graded decision regime (supporting diagnostic). Variants are computational profiles, not clinical diagnoses.

D.2 Alpha sweep

Sweeping α across $\{0.05, 0.1, 0.3, 0.5, 1.0, 2.0, 4.0, 8.0\}$ tests whether precision gain controls reactive social confidence across metrics. β_k movement range increases monotonically with α in the betrayal regime:

Table 7: β_k range by precision-gain α (betrayal regime, 20 seeds).

α	Mean β_k range
0.05	0.147
0.1	0.182
0.3	0.326
0.5	0.416
1.0	0.580
2.0	0.862
4.0	1.051
8.0	1.218

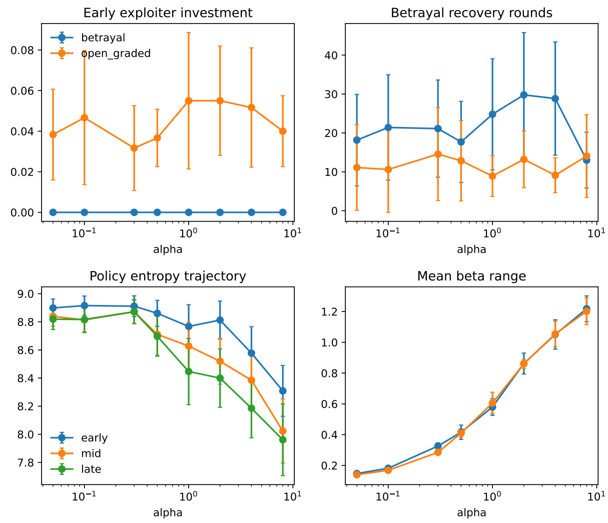

Fig. 6: Precision-gain sweep across open graded and betrayal environments (20 seeds). Higher α expands β_k range and reshapes engagement readouts without monotonic payoff ranking.

D.3 Phenotype quadrants

Crossing prior belief (naive versus cautious) with α (low versus high) defines four behavioral profiles tested in the betrayal environment (20 seeds). Table 8 gives key metrics.

Table 8: Behavioral metrics by phenotype (betrayal environment, 20 seeds). Default agent shown for reference.

Profile	Payoff	Recovery	Gini	β range	Trust asymmetry
Anxious-reactive	1932	26.5	0.338	1.084	0.98
Hypervigilant	1951	20.1	0.423	0.892	1.65
Naive-stubborn	1894	21.3	0.304	0.364	1.19
Avoidant-rigid	1889	14.9	0.357	0.322	1.21
Default	1991	23.0	0.423	0.893	1.68

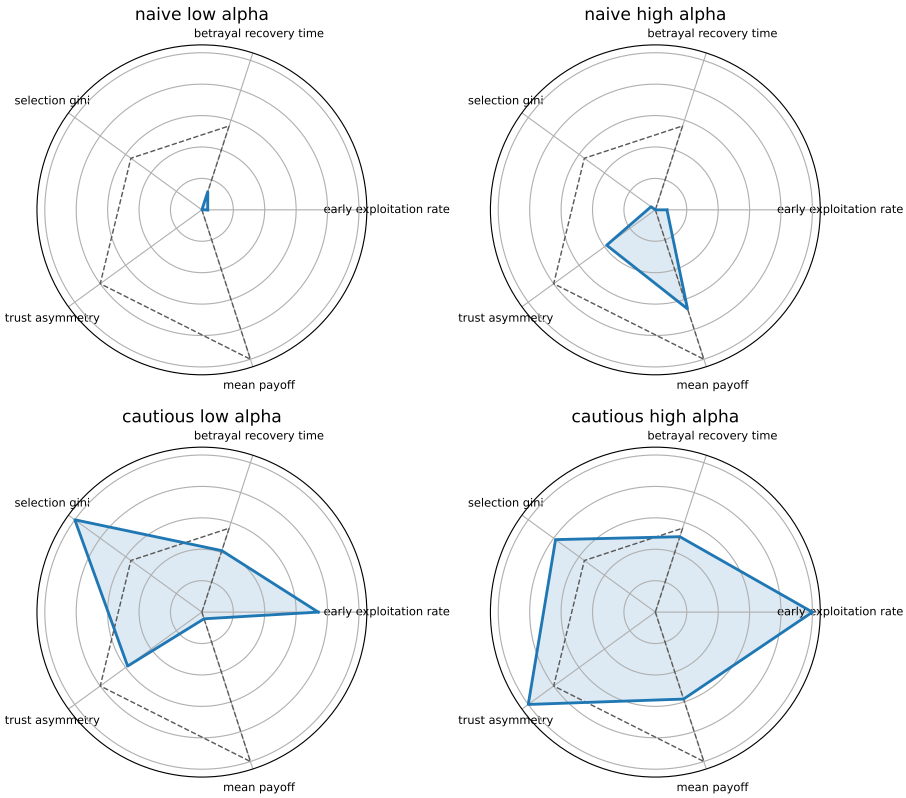

Fig. 7: Phenotype quadrant profiles from the prior-by-α factorial (betrayal environment, 20 seeds). Profiles are computational analogues, not clinical classifications.

D.4 Forgiveness and trust repair

The forgiveness paradigm uses a 200-round partner-choice episode: cooperative baseline (rounds 1–80), betrayal on P0 (rounds 81–120), then reversion to cooperative/trusting behavior (rounds 121–200). Readouts include reengagement latency, post-return selection, payoff recovery, and β_k recovery trajectories for the reverted partner.

[*Forgiveness results pending: report reengagement latency, post-return selection, payoff recovery, and whether high-α controls reengagement speed while cautious priors limit the trust-repair ceiling.*]

D.5 Mixed-volatility extended diagnostics

The primary mixed-volatility table and figure remain in Section 3.6. Extended Exp D diagnostics include per-partner β_k trajectories, rolling P0 selection concentration, and pairwise stability–precision correlations used to compute the discrimination index and P0 false-positive rate reported in the main text.

D.6 Additional robustness figures

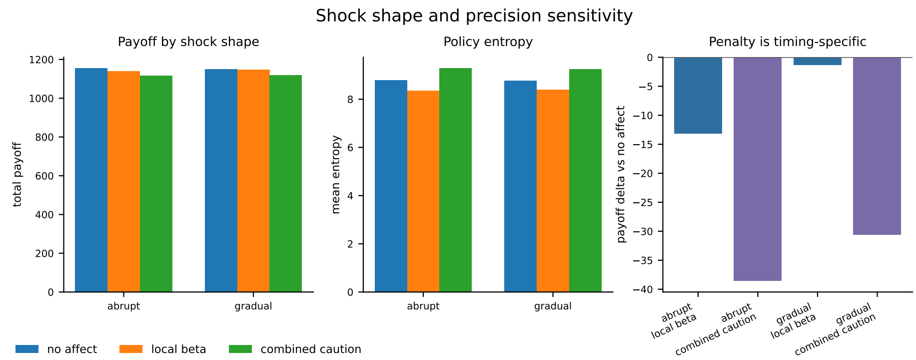

Fig. 8: Shock-shape sensitivity summary (supporting diagnostic). Abrupt versus gradual betrayal schedules under precision-gain and prior perturbations.